Kai Hori

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EDUCATION

University of California, Davis
B.S. ***Mechanical Engineering****, B.S.* ***Aerospace Science and Engineering*** GPA: 3.614/4.0
Relevant Coursework: Mechanics of Materials, Fluid Mechanics, Thermodynamics, Heat Transfer, Automatic Control of Engineering Systems, Aircraft Propulsion, Applied Aerodynamics, Compressible Aerodynamics, Rocket Propulsion, and Space Vehicle Design

TECHNICAL SKILLS

CAD & Engineering Software: Onshape, SOLIDWORKS, Fusion 360, ANSYS Fluent, NASTRAN/PATRAN
Programming & Analysis: MATLAB, Simulink, Python, C#, Fortran, Unity, HLSL
Manufacturing & Prototyping: 3D Printing, Manual Milling, Lathe, CNC Milling, Drilling, Welding
Certifications: Remote Pilot Certificate with a sUAS rating issued by the FAA (Part 107) August 2026
Languages: English (Fluent), Japanese (Fluent)

ENGINEERING AND RESEARCH EXPERIENCE

Mission Operations Intern - ***ispace.inc, S3-1 Lunar Lander Mission Operations*** October 2026 - Present

- Develop Python command-and-telemetry procedures for lunar lander operations, backed by comprehensive unit tests in a CI/CD pipeline to verify them before flight use
- Automate standards-adherence checks and AI-agent workflows that triage and remediate CI/CD test failures, cutting manual review effort
- Collaborate with multidisciplinary engineering teams in spacecraft design reviews to integrate operability requirements

Research Assistant - ***UC Davis Human/Robotics/Vehicle Integration & Performance Laboratory*** Feb 2026 - Present

- Designed, fabricated, and field-developed a satellite ground-station hardware chain from scratch – QFH antenna and parabolic-dish feed assembly – taking each from CAD through machining and assembly to a functional system.
- Conducted iterative field testing and hardware troubleshooting to evaluate mechanical alignment, mounting configuration, antenna positioning, and environmental effects on system performance.

Structure Lead - ***Truss-Based Wing Structural & Aeroelastic Analysis Class Project*** April 2026 -June 2026

- Built finite-element models (NASTRAN/PATRAN) of a truss-based structure; performed static and modal analyses to evaluate structural behavior, natural frequencies, and mode shapes.
- Used MATLAB and flutter-analysis results to assess aeroelastic stability, including flutter and divergence behavior across the operating envelope.

Communications Lead - ***Spacecraft Inspection Mission Design Project, Senior Capstone Class*** April 2026 -June 2026

- Led multidisciplinary engineering analyses and system-level trade studies for a heliocentric spacecraft design, coordinating technical interfaces across communications, power, payload, and spacecraft operations.
- Closed the X-band communications link for a heliocentric orbit (requirements hierarchy, V&V plan, FMEA), verified with FreeFlyer-based performance models and system-level trade studies from PDR to CDR documentation.

Payload Lead - ***Spacecraft Inspection Mission Redesign Project, Senior Capstone Class*** January 2026 - March 2026

- Redesigned NASA's CPOD CubeSat concept for autonomous, non-docking spacecraft inspection, evaluating LiDAR integration, sensor performance, mass, power, and onboard-processing constraints.

Research Intern - ***NeuralX*** July 2025 - September 2025

- Developed real-time GPU-parallelized Smoothed Particle Hydrodynamics simulations in Unity to model water-like fluid behavior with 64,000 particles by writing custom HLSL shaders and optimizing spatial-neighbor search in C#.
- Validated numerical stability of the resulting fluid behavior against expected physical dynamics, iterating on shader logic to fix instabilities introduced by the parallelization and physics.

OTHER WORK EXPERIENCE

Co-Instructor - ***UC Davis International Center*** June 2024 - June 2026

- Co-developed and delivered business curriculum for a non-technical audience, translating finance, operations, and marketing concepts into applications relevant to their agriculture applications – direct experience tailoring technical communication to a target audience.
- Partnered with a lead instructor as a culture intermediary and liaison to build a business course for the Japanese Agricultural Training Program in partnership with the Japanese Agricultural Exchange Council.